

ICS553 Machine Learning Essentials

Prosit 1: Data-Driven Allocation of Insecticide-Treated Nets in Ghana

1. The Problem

Where should Ghana's limited supply of insecticide-treated nets (ITNs) go, and how confident can we be in the district-level malaria estimates behind that decision?

There is no one definitive answer. A good submission should clearly justify four aspects: the estimate employed, the level of uncertainty, how you avoided sampling bias and data leakage, and the districts you prioritised, along with your reasoning. Your grade depends on the rigour, transparency, and fairness of your reasoning, rather than merely having the correct figure.

2. The Scenario

Malaria is one of Africa's most serious health challenges: roughly 95% of global malaria deaths occur in the region, and the WHO World Malaria Report 2024 (December 2024) estimates 263 million cases and 597,000 deaths worldwide in 2023. In Ghana, the National Malaria Elimination Programme (NMEP) has carried out nationwide net distribution, most recently in 2023. However, malaria transmission is still uneven: certain interior districts experience a much higher burden, and rural and border communities are frequently underrepresented in national surveys.

Your role. Your team is an analytics consultancy. A new, limited delivery of ITNs has arrived. Advise the NMEP/Minister's allocation committee on where nets will do the most good and be most fairly distributed, and be ready to defend every number.

3. Deliverables

The "portfolio" is due in Week 3:

1. A reproducible notebook (GitHub repo) — the full pipeline, from cleaning to your leakage audit.
2. A Resource Allocation Proposal — a 15-minute presentation to a mock Ministry of Health panel, with probability heatmaps.
3. A 1-page individual reflection — what you learned and how you used AI, declared per the course AI policy.

4. The Data

Use real data only (no fabrication). The instructor provides a de-identified course package in data/:

File	What's in it	Why it matters
ghana_district_cases.csv	District case counts + covariates (rainfall, temperature, elevation, distance to water, net cover)	Your response (case counts) for fitting distributions
ghana_mis_sample.csv	MIS household subset: cluster, rural, net usage, parasitaemia, sample weight	Net coverage & prevalence; its cluster structure drives the sampling lesson
data_dictionary.md	Column-by-column source, type, units, caveats	Your datasheet (LO7)

Rules: the MIS is licensed health data — treat it as confidential, and do not upload it to any generative AI tool. Declare any AI use (course policy).

5. Your Tasks

Do these in your notebook. Support your points with a number or a simple plot where you can — the goal is a clear, honest argument, not a formal proof.

Theme A — Probability Distributions & Sampling

- A1. Look at the response. Plot and describe district case counts. They are counts, right-skewed, and usually over-dispersed (variance > mean). You don't need a formal test to notice this — the plot and a quick mean-vs-variance check are enough.
- A2. Fit a Poisson model. Use the mean as an estimate of the rate λ and overlay a fitted Poisson on your histogram. The main question is simple: Does a Poisson actually match the shape of your data? (If the MLE derivation is tricky, don't get stuck on it — the insight is the fit, not the algebra.)
- A3. Try a Negative-Binomial (NB) model. When counts are over-dispersed, an NB often fits better. Fit one (scikit-learn or statsmodels is fine), read off its extra over-dispersion parameter, and give a basic comparison against the Poisson. A simple mean–variance check and visual comparison are enough; a Q–Q plot is a nice extra, not a requirement.
- A4. Quantify uncertainty. The MIS is a cluster sample, not a simple random sample, so a plain interval understates the uncertainty. Use bootstrapping (resampling with replacement) to

obtain a rough confidence interval for the prevalence in an undersampled rural district. If a full cluster bootstrap feels complex, start with a simple bootstrap and honestly discuss how ignoring the clusters narrows your interval.

Theme B — Data Preprocessing & Leakage

- B1. Explore and find data gaps. Map raw prevalence by district; shade districts by data availability to reveal under-sampled rural/border areas. Note how much is missing and a plausible reason why.
- B2. Handle missingness. Keep imputation simple (mean/median). The one principle to remember: if you later build a predictive model, fit the imputer on the training set and apply it to the test set.
- B3. Encode categories. One-hot encode genuinely categorical fields. Be aware that target encoding can leak the outcome; a Pipeline/ColumnTransformer fitted on train only is the clean pattern, but treat it as best practice, not a hard requirement.
- B4. Avoid leakage. Use a reasonable split (ideally stratified by region) and run a basic leakage check: name the most obvious leaks in your pipeline (e.g., pre-processing on all the data, or using 2023 information to explain 2022). Give a short explanation of each; a numeric "before/after" demonstration earns credit if you can produce one, but a clear written audit is sufficient.

6. Timeline (3 weeks)

Week	Key activity	Output
1	Entry Event, team formation; lecture on distributions & sampling; environment setup	Starter repo, action plan
2	Milestone 1 (geospatial EDA + data-gap map) + gallery walk; lecture on preprocessing & leakage	Draft notebook
3	Milestone 2 (statistical report: Poisson/NB, confidence intervals, leak audit)	Proposal + portfolio due at the debrief

7. Grading (rubric, 1–4 scale)

Criterion	Weight	High marks look like...
Statistical rigour	25%	Detect over-dispersion; fit Poisson and, where it helps, a Negative-Binomial; compare their fit (mean–variance or a simple diagnostic), and use bootstrapping to put some uncertainty around a prevalence estimate.
Pipeline & leakage	25%	A sensible split (ideally stratified); pre-processing kept train-only; the main leakage risks were identified and explained clearly
Equity & ethics	20%	Identify under-sampled districts; wide CIs drive a fair allocation, not just the mean-best one.
Framing & decision	15%	Clear target, unit, and metric; defensible recommendation
Communication & teamwork	15%	Clear 15-min narrative; heatmaps tell the story; panel questions answered

Reflection is required to pass (no weight): it proves your individual understanding and satisfies the AI-use declaration.

8. Learning Outcomes

This Prosit trains the course outcomes: you frame the problem (LO1), build an end-to-end pipeline (LO2), derive the Poisson/NB logic (LO3), validate statistically (LO4), audit bias and fairness (LO5), communicate the proposal (LO6), and critique the data and WHO figure critically (LO7).

9. Submission Checklist

- ☐ No cell fits, encodes, or scales on pooled train+test data
- ☐ A "Leakage Audit" section names each leak risk + a before/after number
- ☐ Every statistic cites a source; the datasheet is referenced
- ☐ AI-use declaration is filled in
- ☐ The repo runs top-to-bottom from a fresh clone